Fourier Transform

https://homepages.inf.ed.ac.uk/rbf/HIPR2/spatdom.htm

Spatial Domain

- For simplicity, assume that **the image I** being considered is formed by projection from **scene S** (which might be a two- or three-dimensional scene, *etc.*).
- The ***spatial domain*** is the **normal image space**, in which a ***change in position in I directly projects to a change in position in S***. Distances in I (in pixels) correspond to real distances (*e.g.* in meters) in S.
- A related term used in this context is ***spatial frequency***, which refers to the (inverse of the) periodicity with which the image intensity values change. Image features with high spatial frequency (such as edges) are those that change greatly in intensity over short image distances.

Frequency Domain

- The ***frequency domain*** is a space in which each image value at **image position F** represents the **amount that the intensity values in image I** vary over a **specific distance related to F**. In the frequency domain, changes in image position correspond to changes in the spatial frequency, (or the rate at which image intensity values) are changing in the spatial domain image I.
- The spatial frequency domain is interesting because: 1) it may make explicit periodic relationships in the spatial domain, and 2) some image processing operators are more efficient or indeed only practical when applied in the frequency domain.

Fourier Transform

- The Fourier Transform is used to convert images from the spatial domain into the frequency domain and vice-versa.
- The Fourier Transform is an important image processing tool which is used to decompose an image into its sine and cosine components.
- The output of the transformation represents the image in the *Fourier* or frequency domain, while the input image is the spatial domain equivalent.
- The Fourier Transform is used in a wide range of applications, such as image analysis, image filtering, image reconstruction and image compression.

Short Time Fourier Transform (STFT)

Fourier Transform

- Fourier Transform reveals which frequency components are present in a given function.

$$f(x) = \sum_{u=0}^{N-1} F(u) e^{\frac{j2\pi ux}{N}}, \; x = 0, 1, \ldots, N-1 \qquad \text{(inverse DFT)}$$

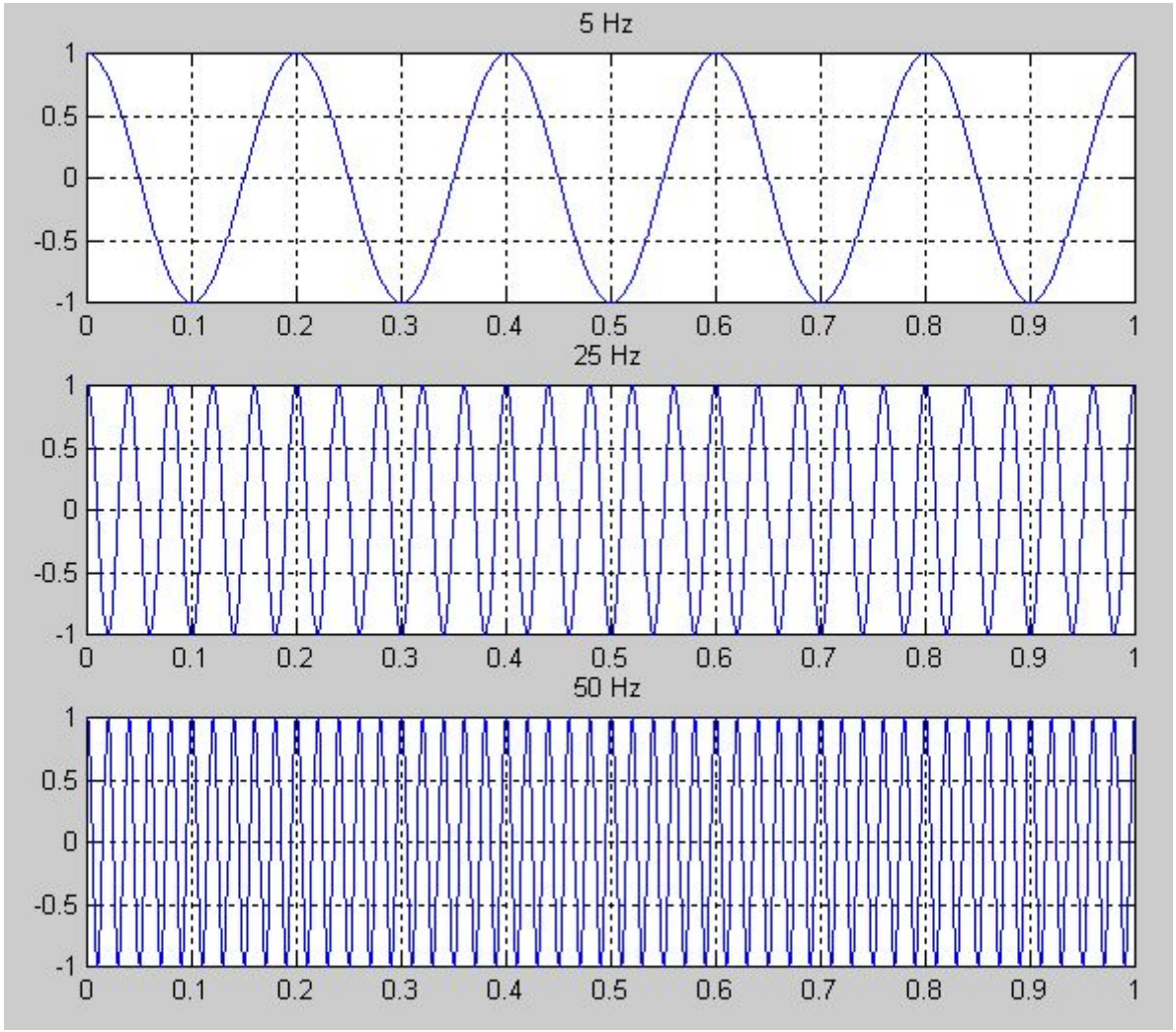
where: $$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{\frac{-j2\pi ux}{N}}, \; u = 0, 1, \ldots, N-1 \qquad \text{(forward DFT)}$$

Examples

$$f_1(t) = \cos(2\pi \cdot 5 \cdot t)$$

$$f_2(t) = \cos(2\pi \cdot 25 \cdot t)$$

$$f_3(t) = \cos(2\pi \cdot 50 \cdot t)$$

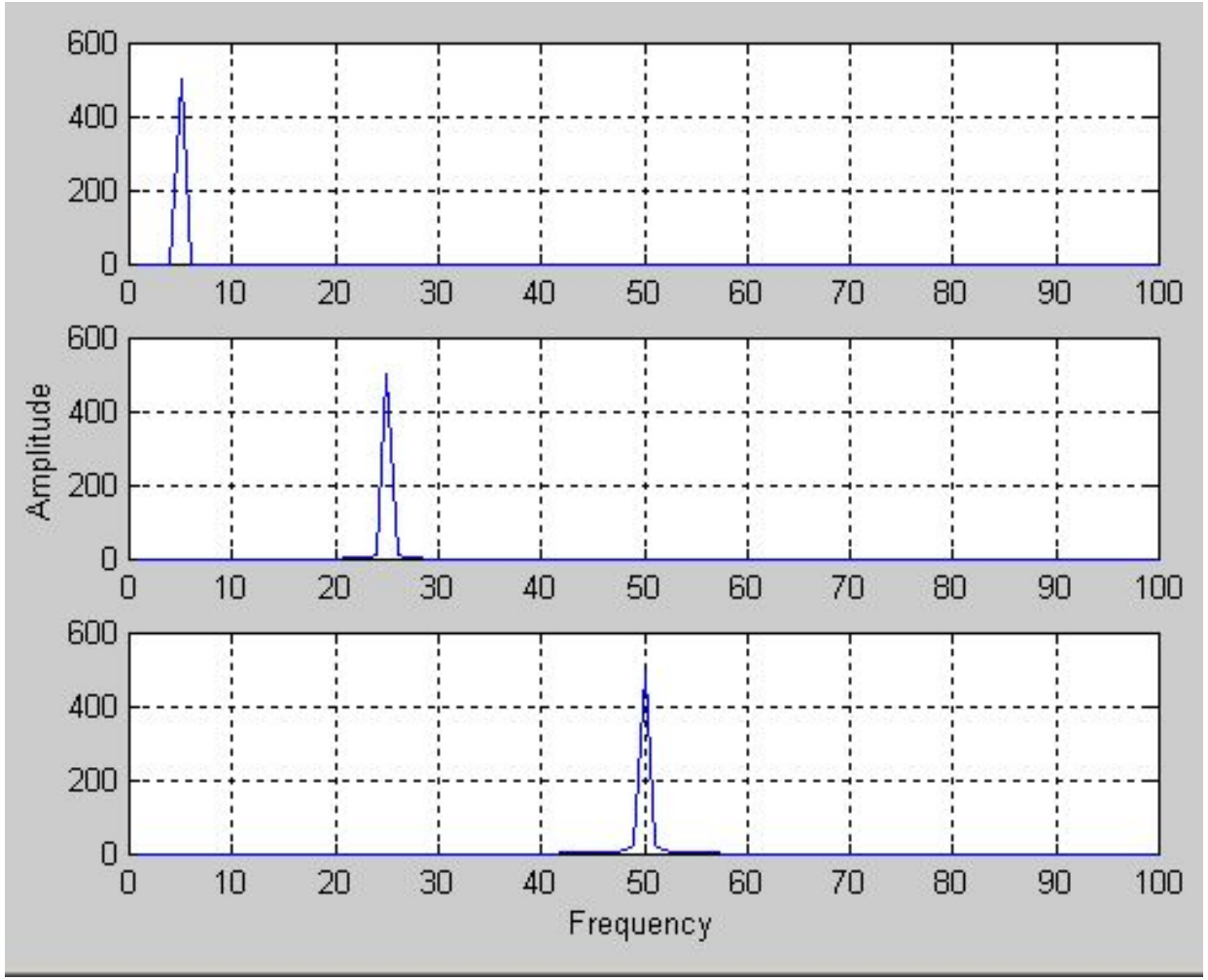


Examples (cont'd)

$F_1(u)$

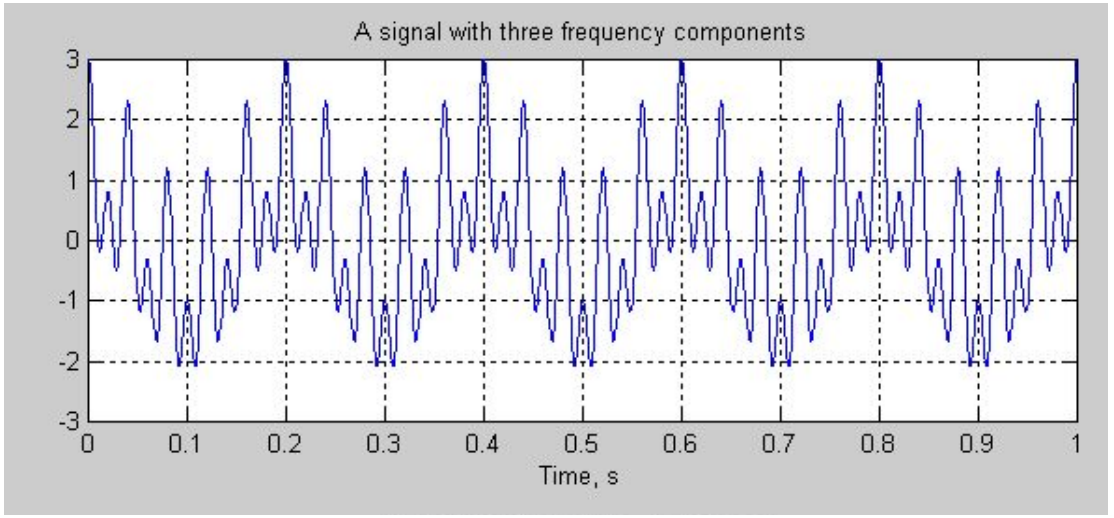
$F_2(u)$

$F_3(u)$

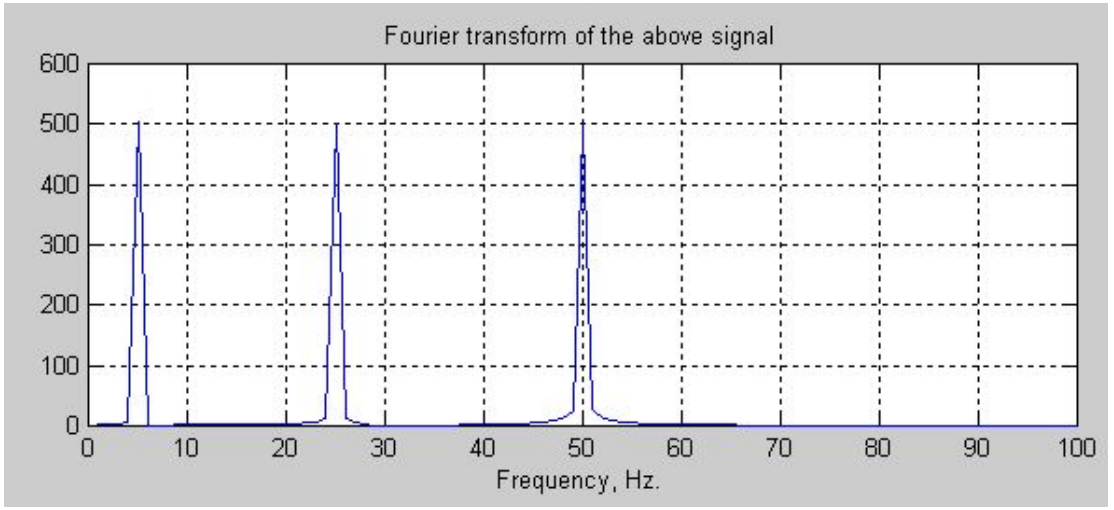


Fourier Analysis – Examples (cont'd)

$$f_4(t) = \cos(2\pi \cdot 5 \cdot t) + \cos(2\pi \cdot 25 \cdot t) + \cos(2\pi \cdot 50 \cdot t)$$

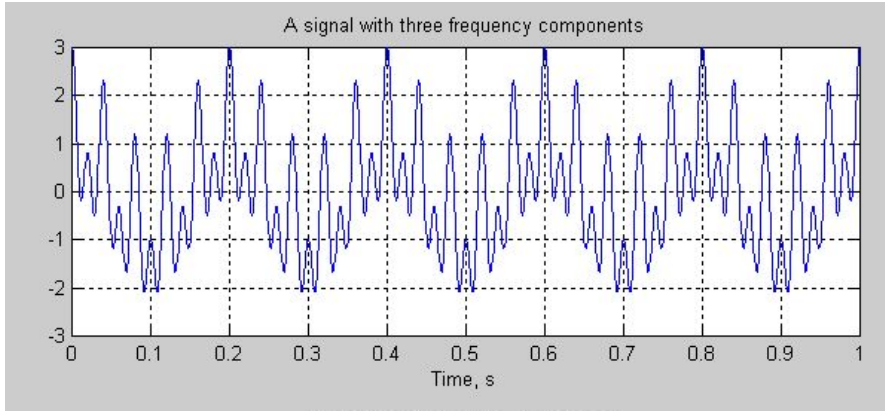


$F_4(u)$?

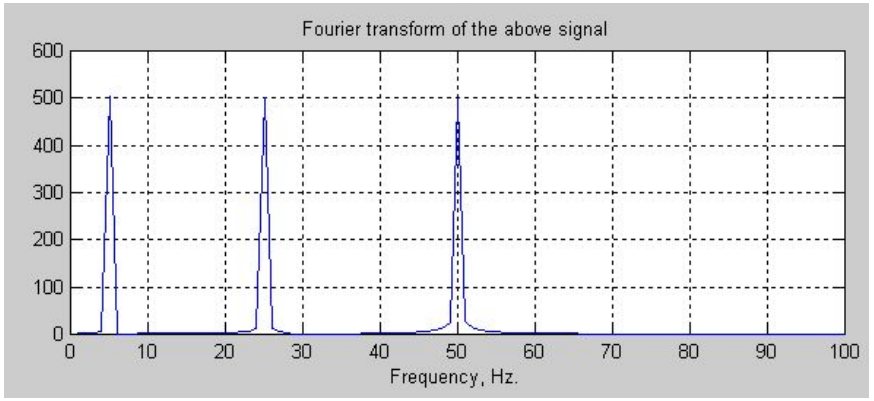


Limitations of Fourier Transform

1. Cannot not provide simultaneous time and frequency localization.



Poor localization in freq domain!

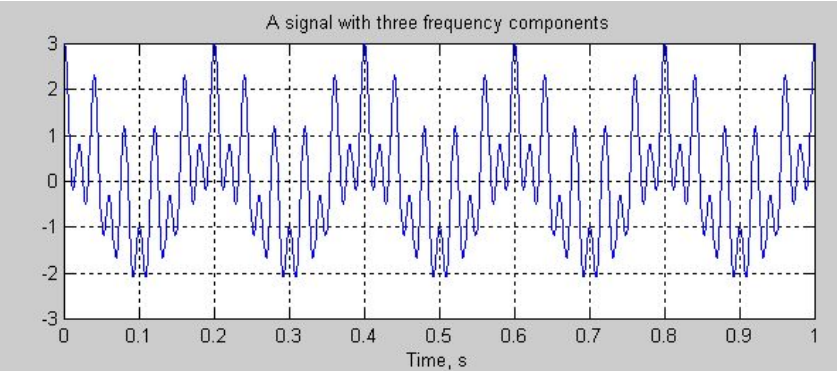


Great localization in freq domain!

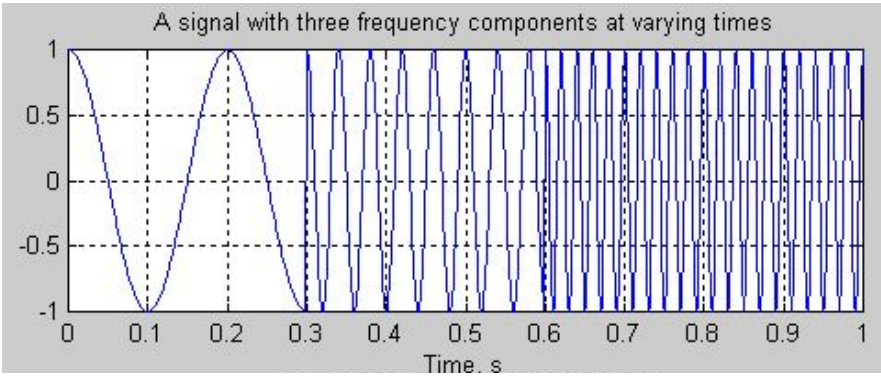
Limitations of Fourier Transform (cont'd)

2. Not very useful for analyzing time-variant, non-stationary signals.

$f_4(t)$ Stationary signal
(non-varying frequency)

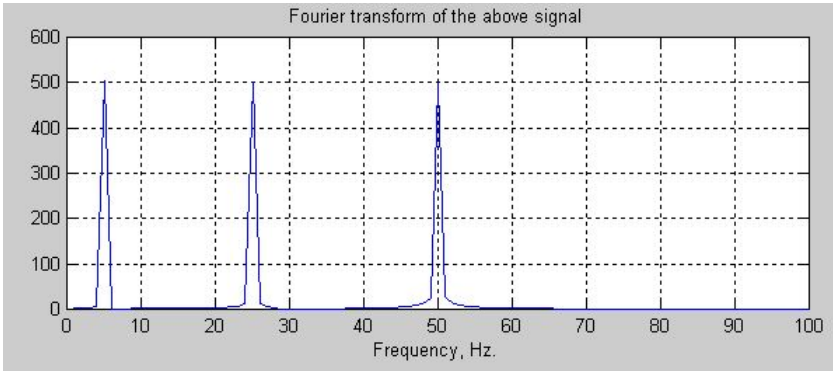


$f_5(t)$ Non-stationary signal
(varying frequency)

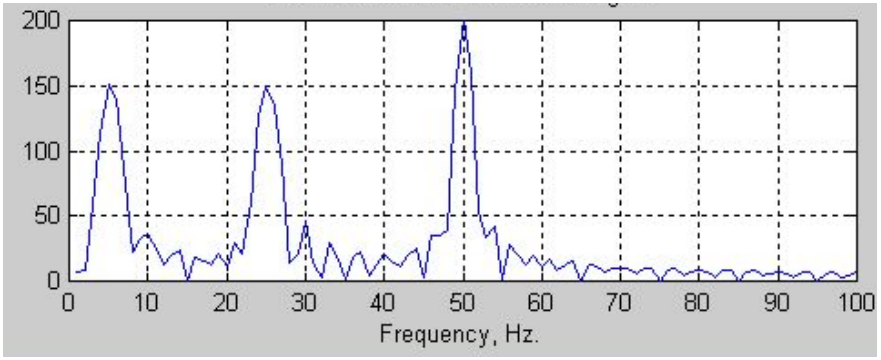


Limitations of Fourier Transform (cont'd)

$F_4(u)$ Three frequency components, present at all times!



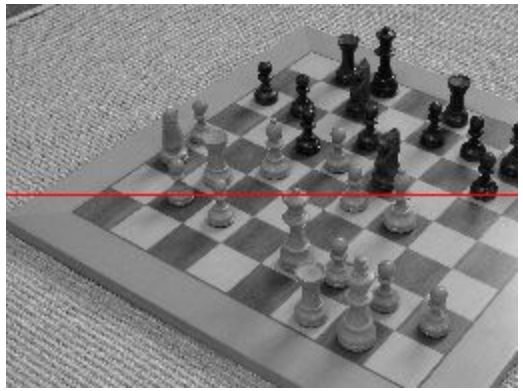
$F_5(u)$ Three frequency components, NOT present at all times!

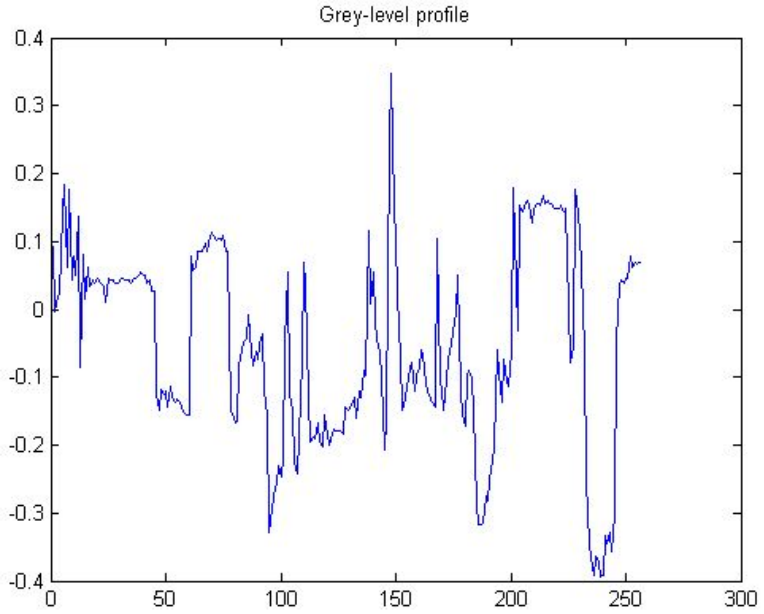


Perfect knowledge of what frequencies exist, but no information about where these frequencies are located in time!

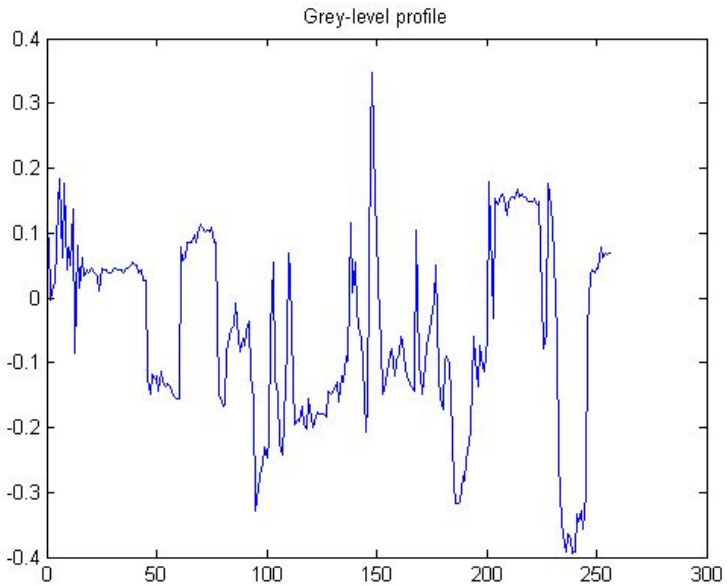
Limitations of Fourier Transform (cont'd)

3. Not efficient for representing non-smooth functions.

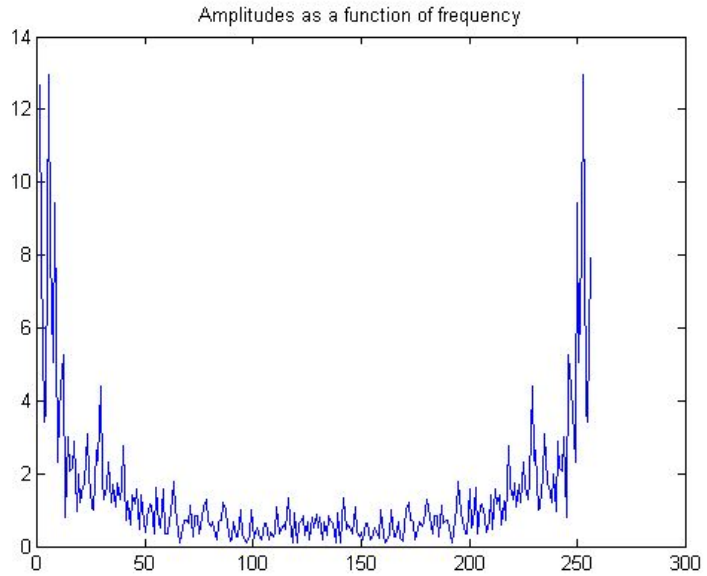




Representing discontinuities or sharp corners (cont'd)



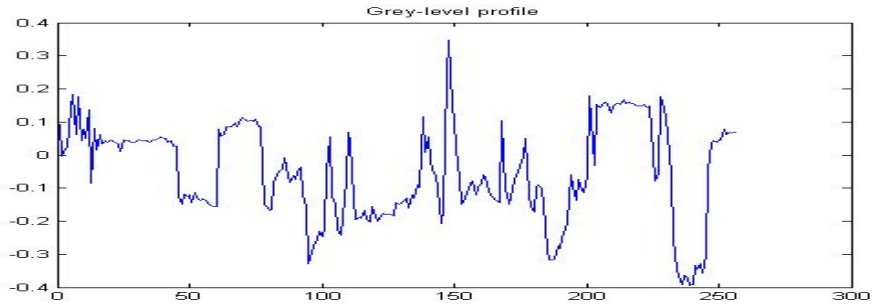
FT

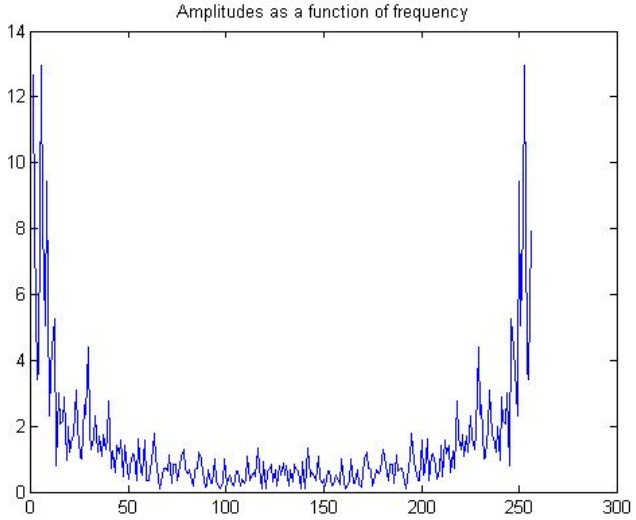


$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) e^{\frac{-j2\pi ux}{N}}, u = 0, 1, \ldots, N-1$$

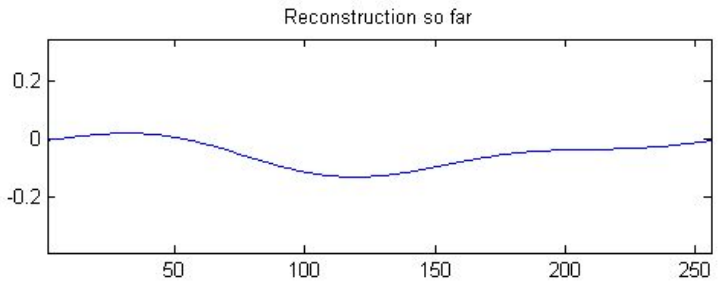
Representing discontinuities or sharp corners (cont'd)

Original





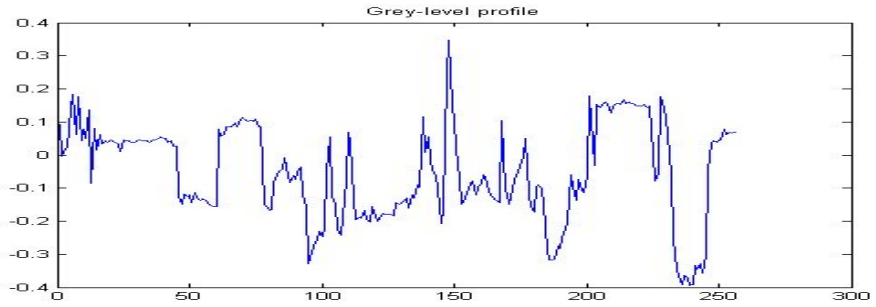
Reconstructed

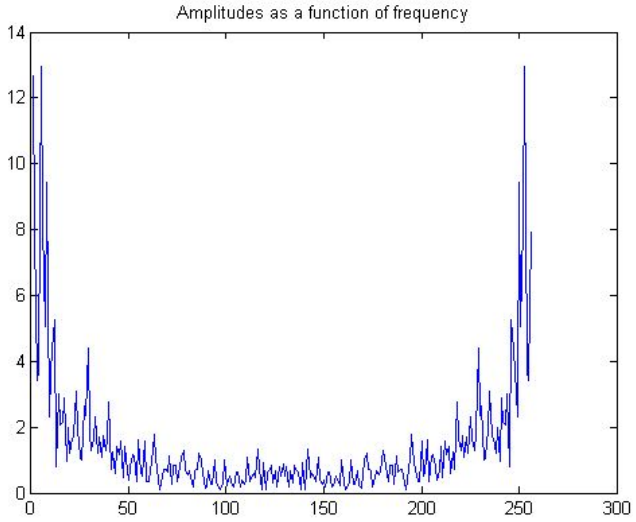


$$f(x) = \sum_{u=0}^{1} F(u) e^{\frac{j2\pi ux}{N}}, \quad x = 0, 1, \ldots, N-1$$

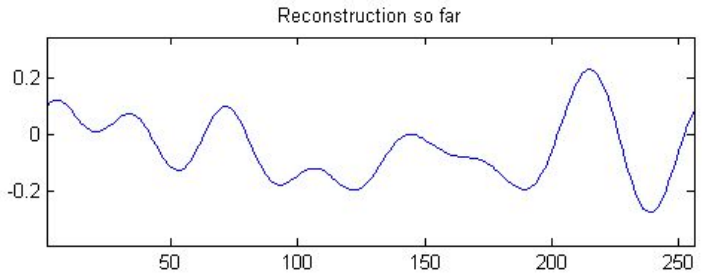
Representing discontinuities or sharp corners (cont'd)

Original





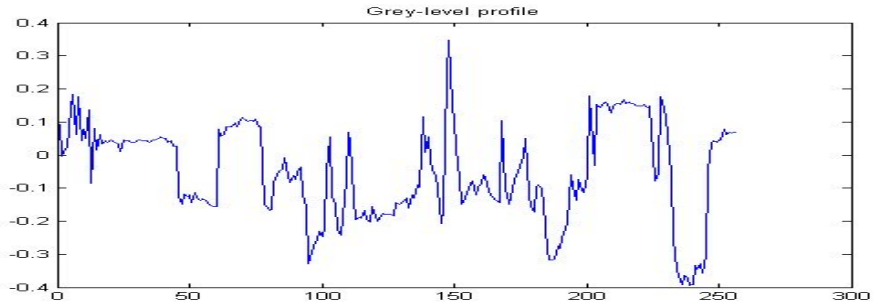
Reconstructed

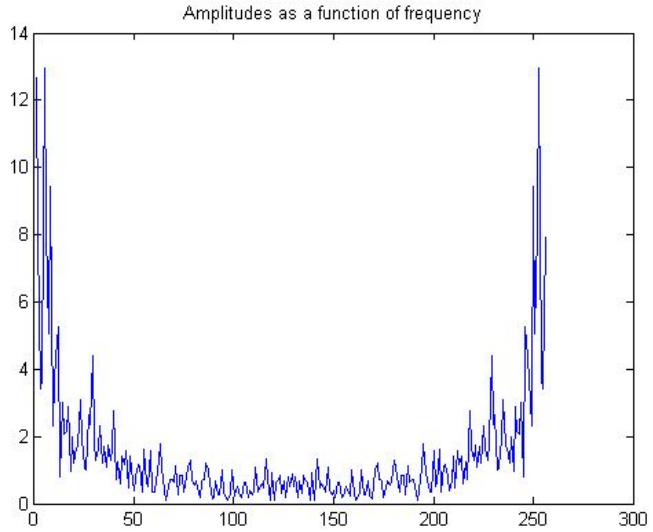


$$f(x) = \sum_{u=0}^{7} F(u) e^{\frac{j2\pi ux}{N}}, \quad x = 0, 1, \ldots, N-1$$

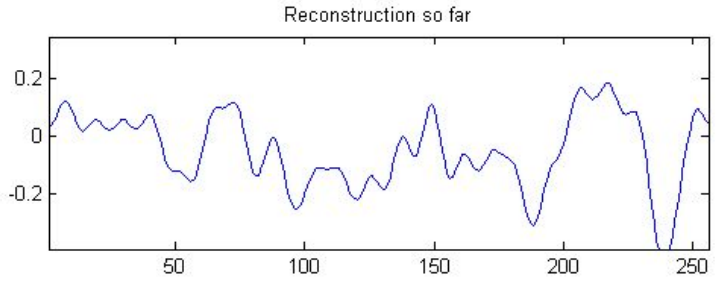
Representing discontinuities or sharp corners (cont'd)

Original





Reconstructed

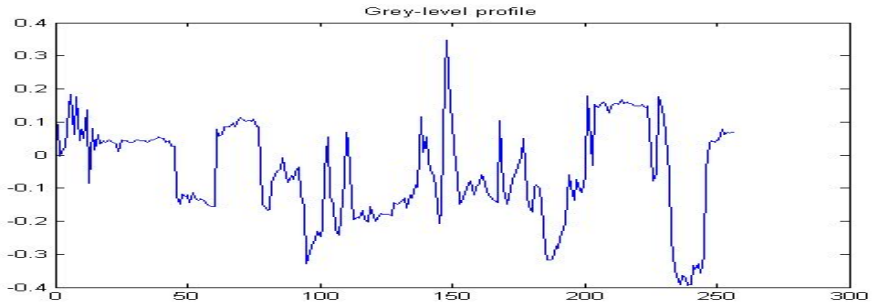


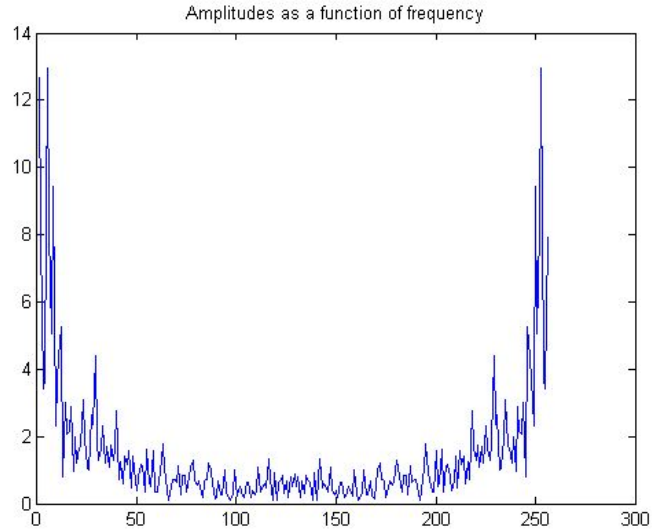
23

$$f(x) = \sum_{u=0} F(u) e^{\frac{j2\pi ux}{N}}, \; x = 0, 1, \ldots, N-1$$

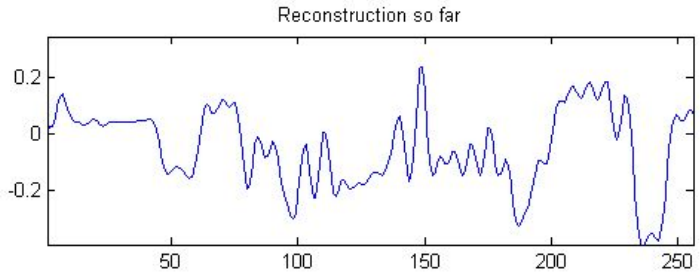
Representing discontinuities or sharp corners (cont'd)

Original





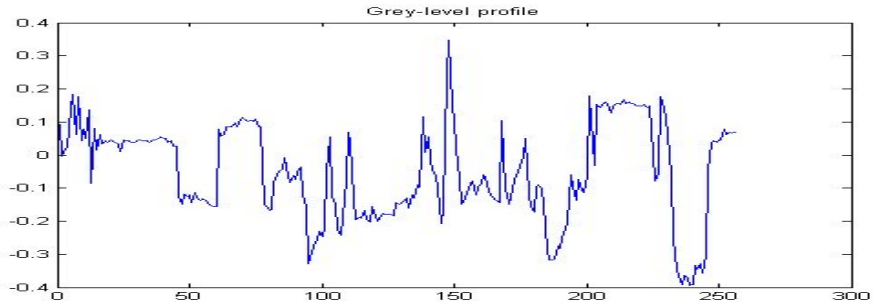
Reconstructed



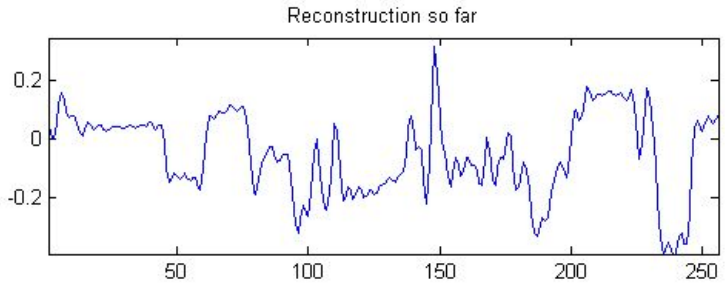
$$f(x) = \sum_{u=0} F(u) e^{\frac{j2\pi ux}{N}}, \; x = 0, 1, \ldots, N-1$$

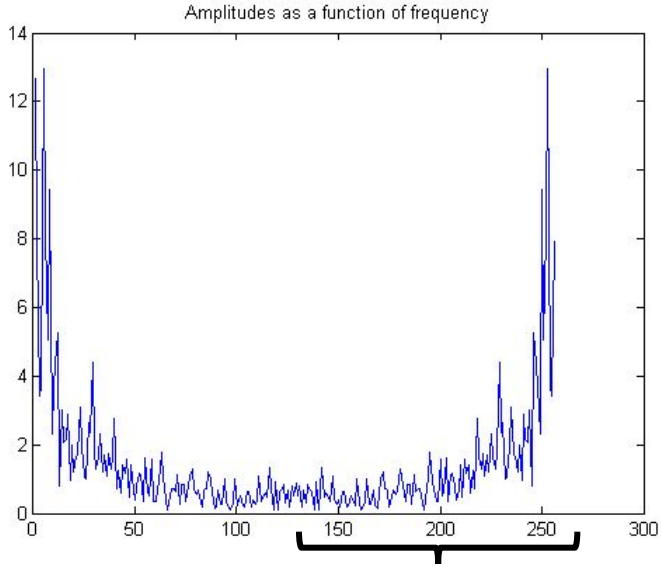
Representing discontinuities or sharp corners (cont'd)

Original



Reconstructed



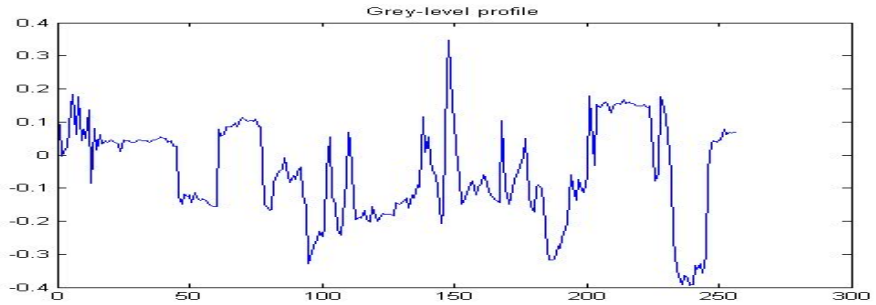


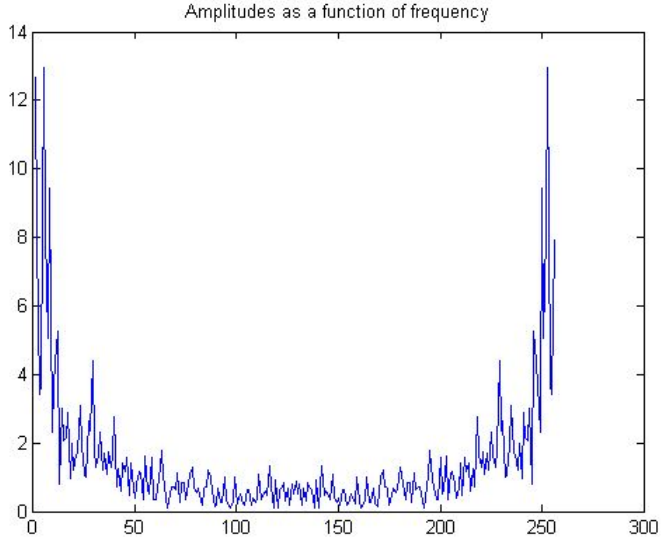
128 coefficients

$$f(x) = \sum_{u=0}^{63} F(u) e^{\frac{j2\pi ux}{N}}, \quad x = 0, 1, \ldots, N-1$$

Representing discontinuities or sharp corners (cont'd)

Original





Reconstructed

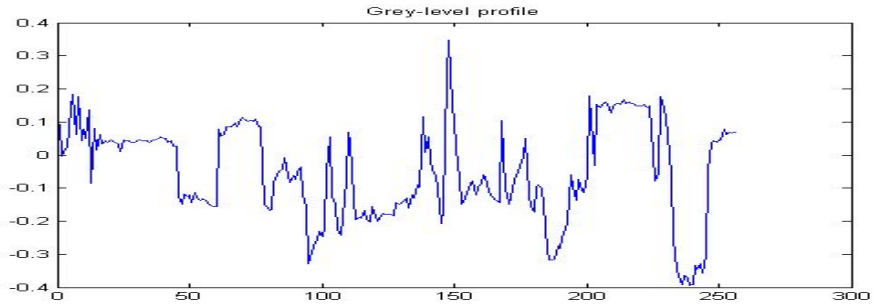


95

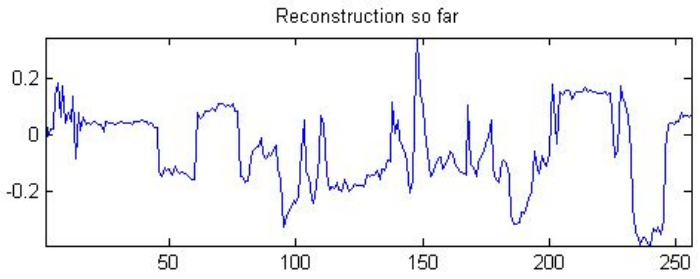
$$f(x) = \sum_{u=0} F(u) e^{\frac{j2\pi ux}{N}}, \; x = 0, 1, \ldots, N-1$$

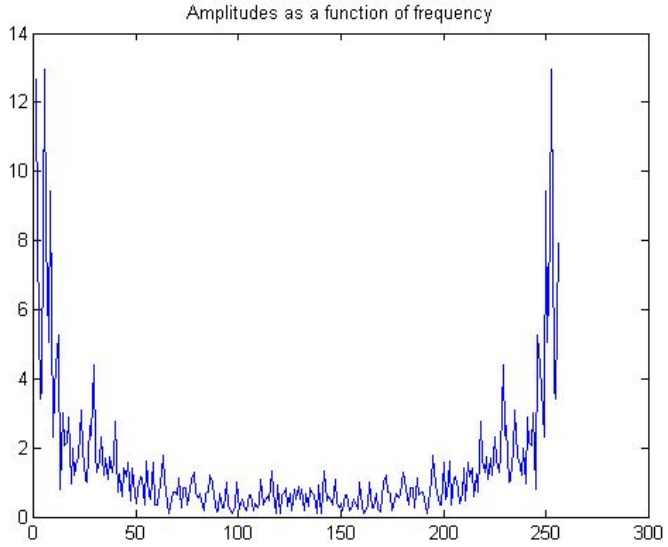
Representing discontinuities or sharp corners (cont'd)

Original



Reconstructed



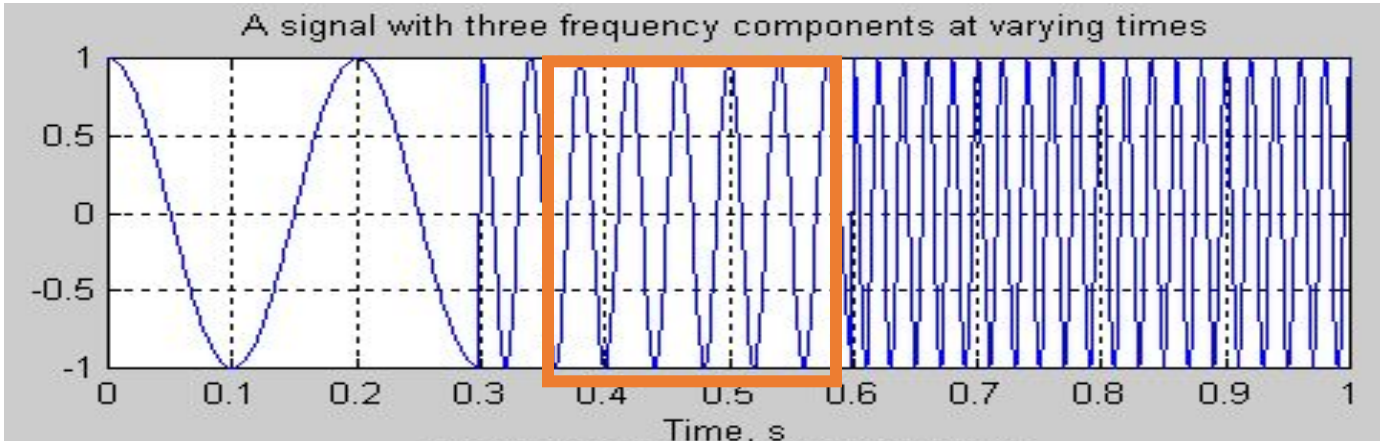


$$f(x) = \sum_{u=0}^{127} F(u) e^{\frac{j2\pi ux}{N}}, \quad x = 0, 1, \ldots, N-1$$

A large number of Fourier components is needed to represent discontinuities.

Short Time Fourier Transform (STFT)

- Segment signal into narrow time intervals (i.e., narrow enough to be considered stationary) and take the FT of each segment.
- Each FT provides the spectral information of a separate time-slice of the signal, providing simultaneous time and frequency information.



STFT - Steps

(1) Choose a window of finite length

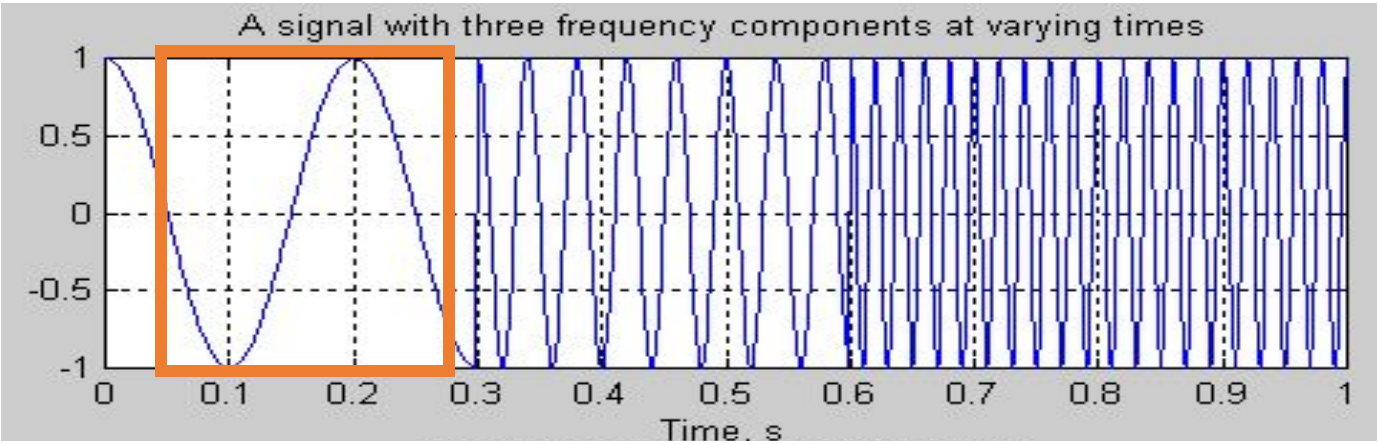
(2) Place the window on top of the signal at t=0

(3) Truncate the signal using this window

(4) Compute the FT of the truncated signal, save results.

(5) Incrementally slide the window to the right

(6) Go to step 3, until window reaches the end of the signal



STFT - Definition

time parameter

frequency parameter

$$STFT_f^u(t',u) = \int_t \left[f(t) \cdot W(t-t') \right] \cdot e^{-j2\pi ut} dt$$

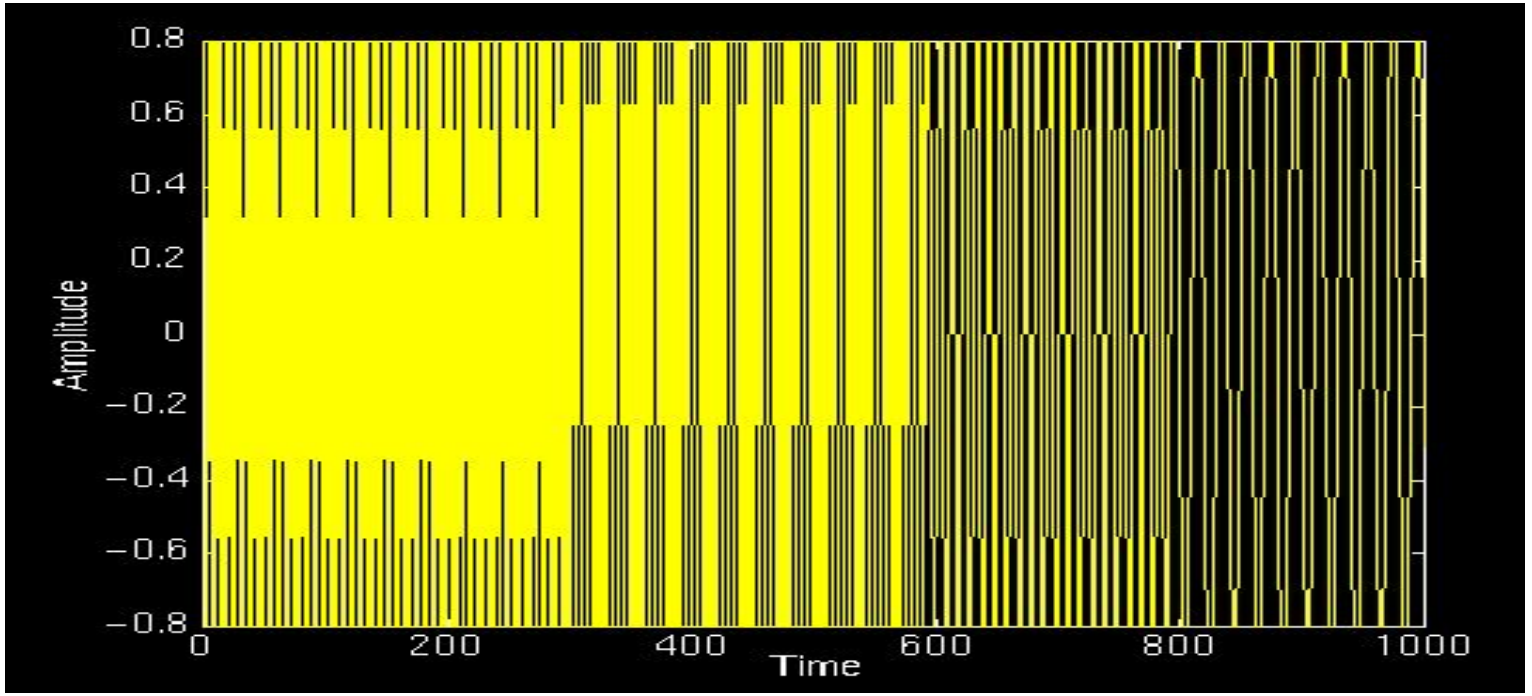
2D function

windowing function

centered at t=t’

Example

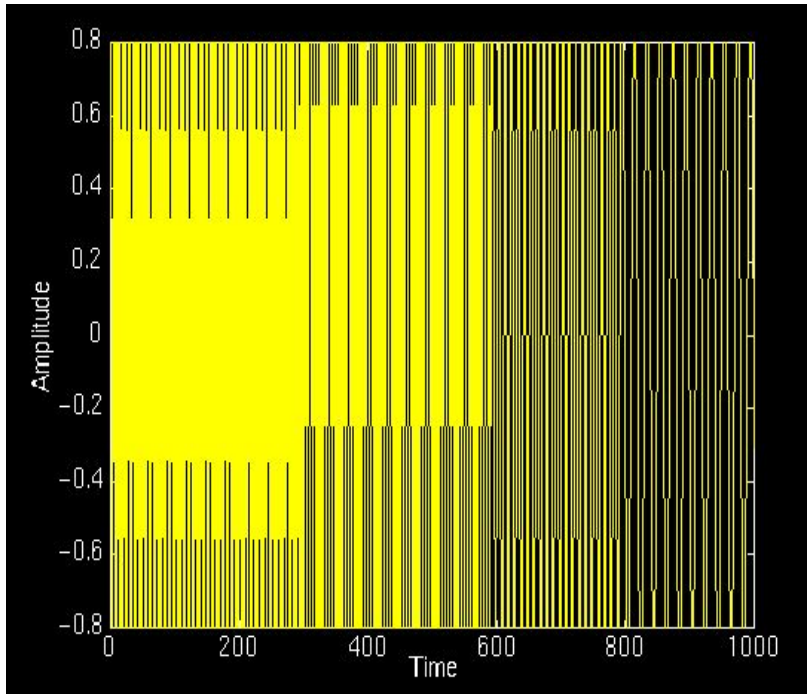
f(t)



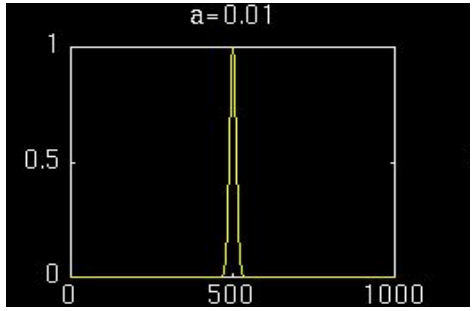
[0 – 300] ms □ 75 Hz sinusoid
[300 – 600] ms □ 50 Hz sinusoid
[600 – 800] ms □ 25 Hz sinusoid
[800 – 1000] ms □10 Hz sinusoid

Example

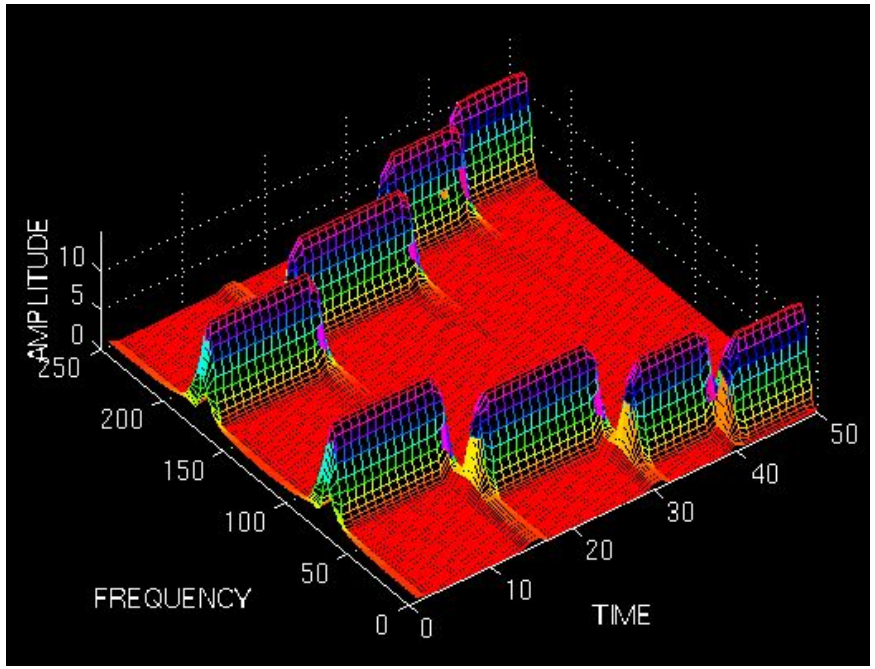
f(t)



[0 – 300] ms □ 75 Hz
[300 – 600] ms □ 50 Hz
[600 – 800] ms □ 25 Hz
[800 – 1000] ms □10 Hz



$$STFT_f^u(t',u)$$



scaled: t/20

Choosing Window W(t)

- What shape should it have?
  - Rectangular, Gaussian, Elliptic …
- How wide should it be?
  - Should be narrow enough to ensure that the portion of the signal falling within the window is stationary.
  - Very narrow windows, however, do not offer good localization in the frequency domain.

STFT Window Size

$$STFT_f^u(t',u) = \int_t \left[f(t) \cdot W(t-t') \right] \cdot e^{-j2\pi ut} dt$$

***W(t)* infinitely long**: $W(t) = u(t)$ □ STFT turns into FT, providing excellent frequency localization, but no time localization.

***W(t)* infinitely short**: $W(t) = \delta(t)$ □ results in the time signal (with a phase factor), providing excellent time localization but no frequency localization.

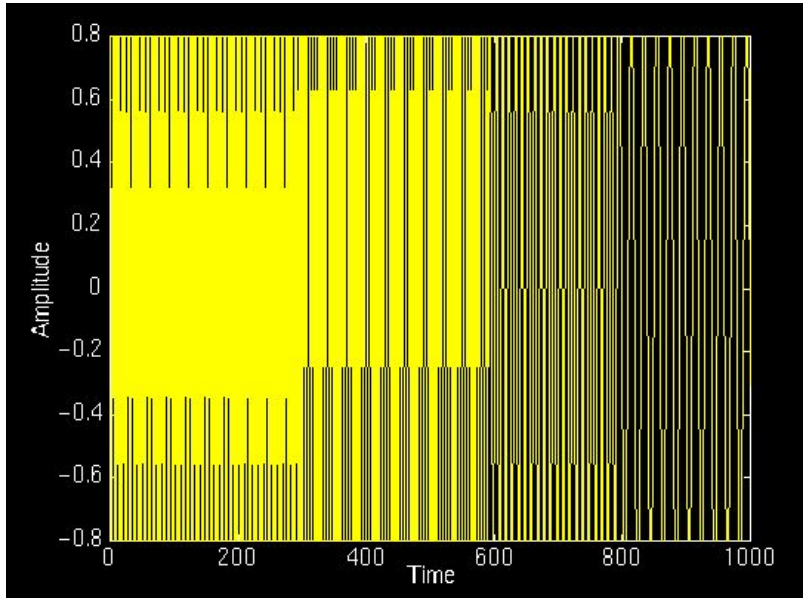
$$STFT_f^u(t',u) = \int_t \left[f(t) \cdot \delta(t-t') \right] \cdot e^{-j2\pi ut} dt = f(t') \cdot e^{-jut'}$$

STFT Window Size (cont’d)

- Wide window good frequency resolution, poor time resolution.

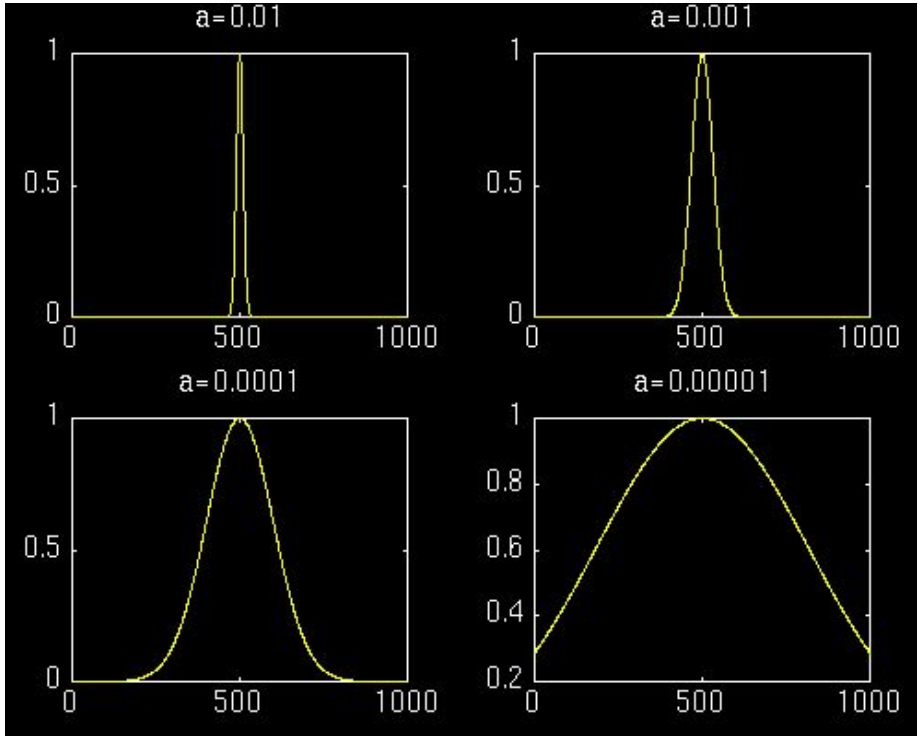
- Narrow window good time resolution, poor frequency resolution.

Example



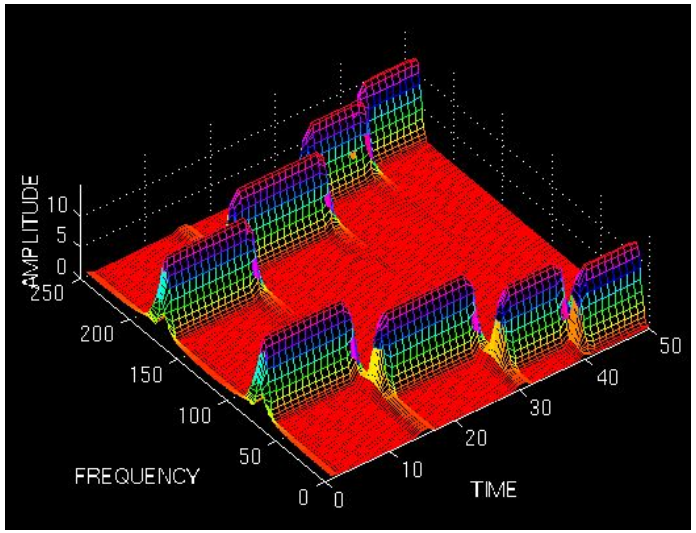
[0 – 300] ms □ 75 Hz
[300 – 600] ms □ 50 Hz
[600 – 800] ms □ 25 Hz
[800 – 1000] ms □10 Hz

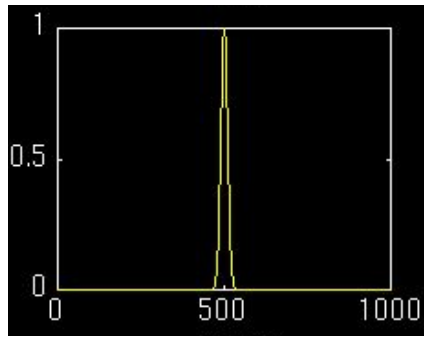
different size windows



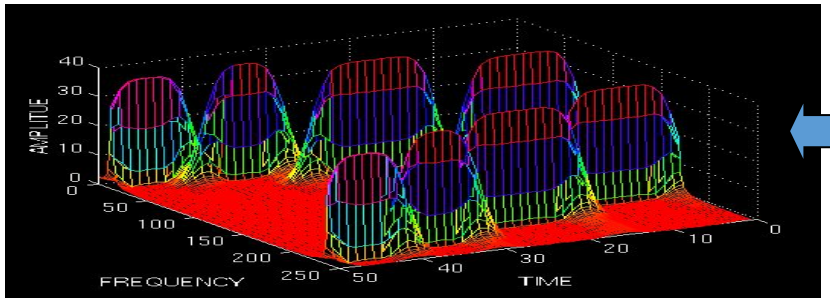
Example (cont'd)

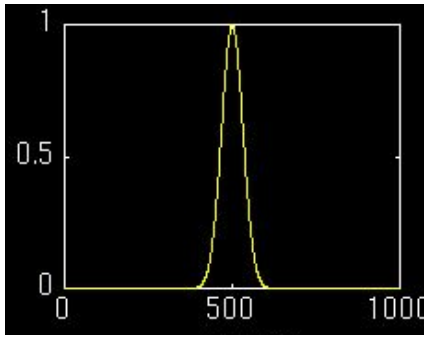
$STFT_f^u(t',u)$





$STFT_f^u(t',u)$

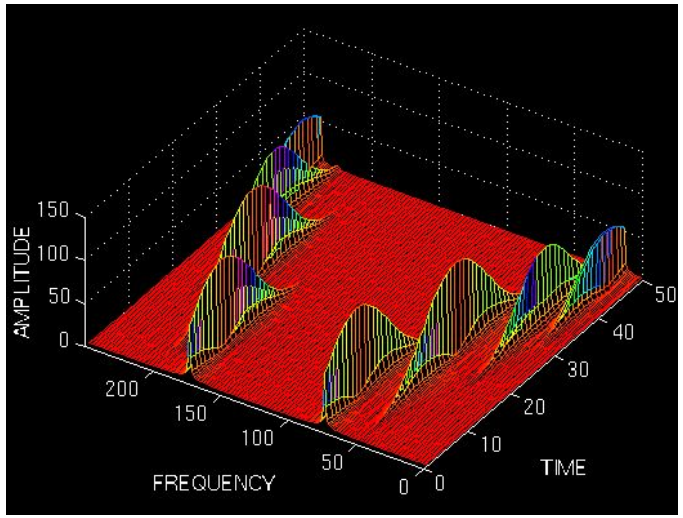


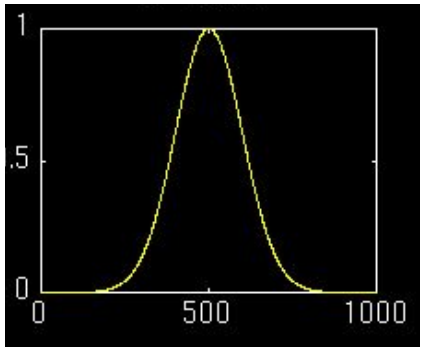


scaled: t/20

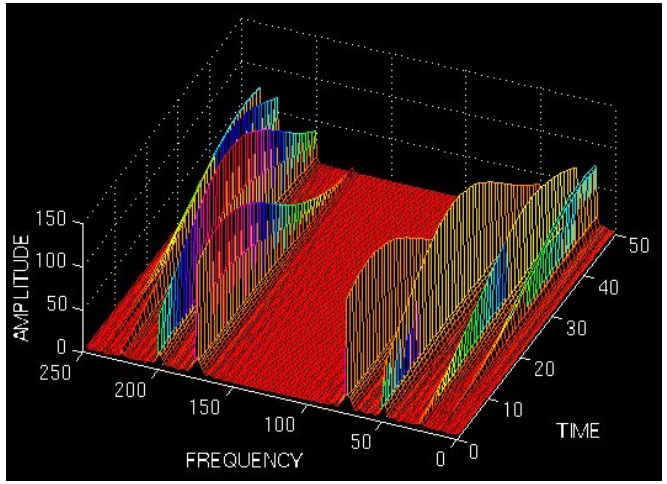
Example (cont'd)

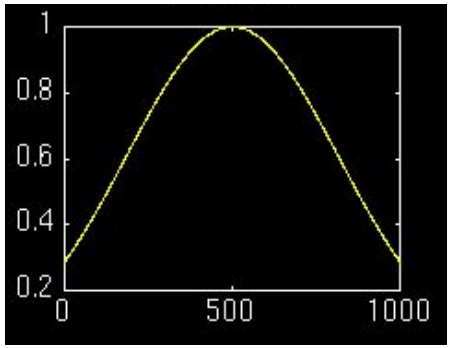
$STFT_f^u(t',u)$





$STFT_f^u(t',u)$





scaled: t/20

Heisenberg (or Uncertainty) Principle

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

Time resolution: How well two spikes in time can be separated from each other in the frequency domain.

Frequency resolution: How well two spectral components can be separated from each other in the time domain

Δt *and* Δf *cannot be made arbitrarily small!*

Heisenberg (or Uncertainty) Principle

- We cannot know the exact time-frequency representation of a signal.
- We can only know what *interval of frequencies* are present in which *time intervals*.